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Why world models fail under intervention: Ontological–causal separation as a necessary structure

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Abstract

Learning world models that can understand, predict, and act within complex physical environments is a fundamental goal of embodied intelligence. However, many dominant approaches—such as latent video models, Dreamer-style imagination agents, and recent JEPA-based predictive architectures—operate on unstructured latent representations, where objects, relations, causality, and intention are implicitly entangled. While effective under observational evaluation, such representations provide limited support for intervention, planning, and counterfactual reasoning. This contrasts sharply with human cognition, which relies on structured reasoning over entities, attributes, and relations to understand and act in the world.

In this work, we introduce LY-GWM (LingYang Graph World Model), a philosophy-inspired, graph-structured world model designed to examine a structural hypothesis: that explicit separation between ontological structure—what entities and relations constitute the world—and causal structure—how these entities evolve under actions—is a necessary condition for coherent world modeling. LY-GWM represents the environment as a dynamic graph composed of entities, attributes, and relations, and models causal dynamics as structured transformations over this graph.

Building on this representation, LY-GWM incorporates three reasoning mechanisms inspired by long-standing philosophical traditions: causal dynamics for action-conditioned state transitions, teleological reasoning for goal-directed behavior, and dialectical novelty detection for identifying structural contradictions and unexplained changes. Through controlled diagnostic environments and high-fidelity humanoid simulation, we show that world models lacking explicit ontological–causal separation can perform well under observation yet systematically fail under intervention, whereas structured graph-based models maintain coherent reasoning across prediction, planning, and counterfactual analysis.

Our results suggest that incorporating explicit ontological and causal structure is not merely a conceptual design choice, but a necessary condition for world models that support intervention-consistent reasoning in embodied intelligence.

1 Introduction

Humans acquire an understanding of the physical world not by memorizing pixels or statistics, but by constructing structured mental models of entities, their attributes, and the relations that govern their interactions. Such models support not only prediction, but also intervention, counterfactual reasoning, intentional action, and curiosity-driven exploration—abilities that emerge early in human

development, long before language mastery or formal instruction. Building artificial world models that exhibit similar structural competence remains a central challenge for embodied intelligence.

Recent progress in self-supervised learning, representation learning, and latent dynamics modeling has led to impressive advances in world modeling, including Dreamer-style imagination agents, latent video predictors, Transformer-based sequence models, and JEPA-style predictive architectures. These systems demonstrate strong predictive performance under observational settings. However, they typically operate on unstructured latent vectors in which object identity, relational structure, and causal dynamics are implicitly entangled. While effective for short-horizon prediction, such representations provide limited interpretability, struggle with systematic generalization, and often fail when agents must reason about interventions rather than observations.

This gap highlights a fundamental limitation of current world models: accurate prediction under observation does not guarantee coherent reasoning under intervention. In particular, when an agent alters the environment through action, a world model must distinguish what entities and relations constitute the world from how those entities change as a consequence of intervention. When ontological representation and causal dynamics are conflated, counterfactual reasoning becomes ill-defined, and planning reduces to correlation-driven extrapolation.

We argue that effective world models must therefore be structured—not merely learned—and that explicit ontological and causal representations are necessary conditions for intervention-consistent reasoning. Philosophical frameworks, long used to characterize human cognition, provide a principled lens for identifying such structural requirements. Based on this perspective, we introduce LY-GWM (LingYang Graph World Model), a philosophy-inspired structured world model designed for embodied humanoid agents.

LY-GWM represents the environment as a dynamic graph composed of three ontological primitives: entities (nodes), attributes (latent properties), and relations (structural and functional edges). On top of this representation, the model incorporates three reasoning mechanisms grounded in philosophical traditions: (1) causal dynamics, which model how graph states evolve under actions and interventions; (2) teleological policy reasoning, which treats goals as explicit computational objects guiding action selection; and (3) dialectical novelty discovery, which identifies structural inconsistencies or unexpected transitions to drive exploration and learning.

Together, these mechanisms enable LY-GWM to function not merely as a predictive engine, but as a structured world model that supports relational reasoning, intervention-aware planning, and interpretable failure detection. We evaluate LY-GWM in controlled diagnostic environments and in a high-fidelity simulation of the LingYang-T3 humanoid robot. The results show that models with explicit ontological-causal structure maintain coherent reasoning under intervention, while latent world models—despite strong observational performance—exhibit systematic failure modes. These findings suggest that integrating philosophical structure into world models is not only conceptually motivated, but necessary for building robust and generalizable embodied intelligence.

2 Related Work

2.1 World Models and Latent Dynamics

Early world models such as PlaNet and Dreamer demonstrated that compact latent representations are sufficient for reinforcement learning and imagination-based control. Subsequent variants including DreamerV2/V3, TD-MPC, latent video predictors, and Transformer-based latent dynamics models improved sample efficiency and prediction quality through larger sequence models and richer self-supervised objectives. Recent predictive architectures such as V-JEPA, RvS, and generative agents like Genie further extended world modeling toward large-scale pretrained systems. However, these approaches rely primarily on vectorized latent states that entangle objects, relations, and causal structure, limiting interpretability and systematic generalization. Our work differs by introducing a structured, graph-based ontology and philosophical reasoning modules, enabling more explicit and compositional representations.

85 2.2 Object-Centric and Relational Representations

86 Object-centric models attempt to factorize scenes into entities through slot attention, MONet, IO-
87 DINE, G-SWM, and scene-structured VAEs. Relational reasoning methods such as Relation Net-
88 works, Graph Neural Networks, and graph-based dynamics models further support modeling inter-
89 actions between objects. While effective for scene decomposition and interaction prediction, these
90 models typically lack mechanisms for goal reasoning, causal abstraction, or purposeful action. LY-
91 GWM builds upon this line of work by integrating ontology-grounded entities and relations with
92 philosophically motivated reasoning modules for planning and exploration.

93 2.3 Causal Learning and Counterfactual Prediction

94 Causal representation learning aims to discover latent structure that supports intervention, coun-
95 terfactual reasoning, and long-horizon prediction. Approaches such as CausalVAE, SCM-based
96 dynamics models, and causal discovery frameworks capture some aspects of causal structure but
97 are not tailored to embodied agents that must act within a graph-evolving environment. In contrast,
98 LY-GWM introduces a Causal Dynamics module that explicitly models how graph states transform
99 under actions, bridging causal abstraction with agent-centric control.

100 2.4 Goal-Directed and Teleological Reasoning

101 Goal-conditioned policies, hierarchical reinforcement learning, and temporal abstraction architec-
102 tures (e.g., options, HAC, skill chaining) provide mechanisms for multi-step behavior. Large lan-
103 guage models have recently been used to express high-level goals and abstract plans for embodied
104 agents. However, these approaches often treat goals as additional inputs rather than ontological
105 objects with explicit structure. Inspired by teleological perspectives in philosophy, LY-GWM for-
106 mulates goals as first-class computational entities within the world model, enabling purpose-aware
107 planning and interpretation.

108 2.5 Curiosity, Exploration, and Novelty Detection

109 Curiosity-driven exploration methods include intrinsic motivation, prediction error, Random Net-
110 work Distillation, empowerment, and surprise-based exploration. Recent work in active world mod-
111 eling encourages agents to explore trajectories that reduce uncertainty or maximize information gain.
112 Nevertheless, these techniques do not provide a framework for detecting contradictions or structural
113 inconsistencies in the world model itself. The Dialectical Novelty Discovery module in LY-GWM
114 introduces a mechanism inspired by dialectical reasoning to identify unexpected or contradictory
115 transitions, creating a more cognitively grounded exploration strategy.

116 2.6 Embodied Agents and Humanoid Control

117 Embodied learning frameworks such as MuJoCo-based RL agents, ControlNet-style policies,
118 Whole-Body Control, and large-scale humanoid controllers (e.g., OpenAI’s Humanoid, Figure 01,
119 Unitree G1 models) focus primarily on control and locomotion rather than structured cognition.
120 Our work complements these efforts by providing a high-level world model that supports relational
121 reasoning, causal prediction, and goal-driven planning for humanoid agents such as LingYang-T3.

122 3 LY-GWM Framework

123 We propose LY-GWM (LingYang Graph World Model), a philosophy-inspired structured world
124 model for humanoid robots. LY-GWM consists of two core components: (1) an ontological represen-
125 tation based on entities, attributes, and relations, and (2) three reasoning modules: causal dynamics,
126 teleological policy reasoning, and dialectical novelty discovery. Together, these components form
127 a unified framework for structured scene understanding, causal prediction, goal-oriented planning,
128 and curiosity-driven exploration.

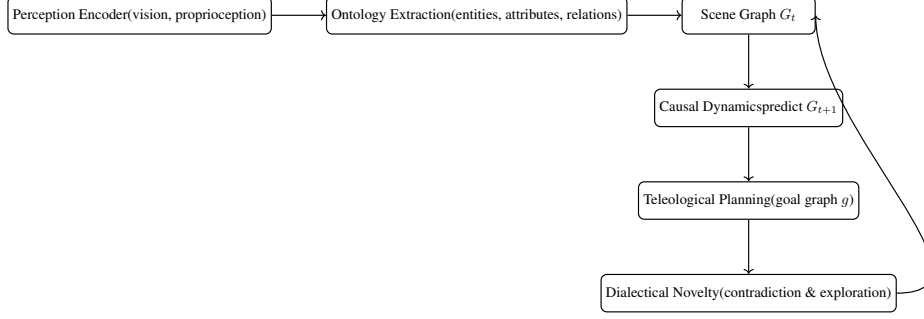


Figure 1: Overall architecture of LY-GWM.

3.1 Ontological Structure: Entities, Attributes, and Relations

The core representation of LY-GWM is a dynamic graph:

$$G_t = (V_t, A_t, R_t)$$

where $V_t = \{v_1, \dots, v_N\}$ represents entities in the environment, A_t represents latent attributes associated with each entity, and R_t encodes spatial, functional, or causal relations between entities.

Unlike traditional world models that compress all information into a dense latent vector, LY-GWM explicitly decomposes the world into structured components. This decomposition enhances interpretability, compositionality, and systematic generalization, enabling more robust reasoning under long-horizon and relational tasks.

3.2 Causal Dynamics Module

The causal dynamics module models how actions induce changes in the graph structure. Formally:

$$G_{t+1} = f_{\text{causal}}(G_t, a_t)$$

where a_t is the agent’s action. This module captures object motion, attribute transitions, and relation updates, inspired by the Aristotelian notion of efficient cause.

By predicting changes directly at the graph level rather than in an unstructured latent vector, LY-GWM enables counterfactual reasoning, intervention-based prediction, and causal abstraction—capabilities that remain challenging for existing vector-based world models.

3.3 Teleological Policy Reasoning

Teleology concerns reasoning about actions in terms of goals or purposes. In LY-GWM, a goal is represented as a structured graph:

$$g = (V_g, A_g, R_g)$$

The agent selects actions that move the current world graph closer to the goal graph. Formally:

$$\pi_{\text{tele}}(a_t \mid G_t, g) = \arg \min_{a_t} \mathcal{D}(G_{t+1}, g)$$

where $\mathcal{D}$ measures structural distance between two graphs.

This provides interpretable, goal-consistent behavior that differs from conventional reward-maximizing reinforcement learning. Teleological reasoning supports planning, symbolic alignment, and human-like intentionality.

3.4 Dialectical Novelty Discovery

Dialectical philosophy identifies contradiction as the source of change and knowledge growth. Based on this idea, LY-GWM defines novelty as structural inconsistency between predicted and observed graph states:

$$n_t = \mathcal{C}(G_t, G_{t+1})$$

where $\mathcal{C}$ quantifies unexpected attribute changes, new relations, or deviations from predicted dynamics.

High novelty automatically triggers exploration behaviors, enabling the agent to discover new object affordances, unseen relations, or overlooked dynamics. This goes beyond simple prediction-error curiosity by capturing deeper structural contradictions.

3.5 Unified World Model Architecture

LY-GWM integrates six modules into a coherent cognitive loop:

- Perception Encoder: extracts spatial and visual features.
- Graph Constructor: builds the entity-attribute-relation graph.
- Causal Dynamics Module: predicts graph transitions.
- Teleological Policy Module: selects goal-directed actions.
- Dialectical Module: detects contradictions to drive exploration.
- Graph Updater: produces the next graph state G_{t+1} .

This architecture forms a new paradigm for embodied intelligence: from structured representation to philosophically inspired reasoning to action generation. LY-GWM provides a foundation for interpretable prediction, relational reasoning, and goal-oriented control in humanoid robotic systems.

4 Method

In this section, we present the mathematical formulation of LY-GWM. The model consists of a structured graph representation, a causal state transition function, a teleological policy objective, a dialectical novelty mechanism, and a unified training framework. Together, these components form a coherent world model capable of relational reasoning, causal prediction, and goal-directed control for humanoid robots.

4.1 Graph Representation

At time t , the world state is represented as a dynamic graph:

$$G_t = (V_t, A_t, R_t)$$

where $V_t = \{v_1, \dots, v_N\}$ is the set of entities, $A_t = \{a_1, \dots, a_N\}$ denotes their latent attributes, and $R_t = \{r_{ij}\}$ encodes directed or undirected relations between entity pairs.

Each entity v_i is parameterized by:

$$h_i = \phi_{\text{enc}}(o_i)$$

where o_i is the perceptual observation (e.g., visual crop, depth feature) and ϕ_{enc} is a learned encoder. Attributes and relations are updated jointly through a graph message-passing network:

$$h'_i = \psi_h \left(h_i, \sum_{j \neq i} \psi_r(h_i, h_j, r_{ij}) \right)$$

4.2 Causal State Transition

Given an action a_t , LY-GWM predicts the next graph state by applying a causal transition operator:

$$G_{t+1} = f_{\text{causal}}(G_t, a_t)$$

This operator consists of two components:

Entity Dynamics.

$$h_i^{t+1} = \psi_{\text{dyn}}(h_i^t, a_t, c_i^t)$$

where c_i^t is the aggregated relational context.

Relation Dynamics.

$$r_{ij}^{t+1} = \psi_{\text{rel}}(h_i^t, h_j^t, r_{ij}^t, a_t)$$

189 This structure enables intervention-based reasoning and counterfactual prediction.

190 4.3 Teleological Policy Objective

191 A goal is represented as a target graph:

$$g = (V_g, A_g, R_g)$$

192 To select an action, LY-GWM minimizes the structural distance between predicted future graphs and
193 the goal graph:

$$\pi_{\text{tele}}(a_t \mid G_t, g) = \arg \min_{a_t} \mathcal{D}(f_{\text{causal}}(G_t, a_t), g)$$

194 where $\mathcal{D}$ is a differentiable graph-structure distance:

$$\mathcal{D}(G_1, G_2) = \sum_i \|h_i^{(1)} - h_i^{(2)}\|_2^2 + \lambda \sum_{i,j} \|r_{ij}^{(1)} - r_{ij}^{(2)}\|_2^2$$

195 This formulation provides interpretable goal-directed behavior.

196 4.4 Dialectical Novelty Mechanism

197 To detect contradictions or unexpected transitions, we define a novelty signal:

$$n_t = \mathcal{C}(G_t, \hat{G}_t)$$

198 where $\hat{G}_t = f_{\text{causal}}(G_{t-1}, a_{t-1})$ is the predicted graph.

199 The contradiction measure is:

$$\mathcal{C}(G_1, G_2) = \sum_i \|h_i^{(1)} - h_i^{(2)}\|_1 + \gamma \sum_{i,j} \|r_{ij}^{(1)} - r_{ij}^{(2)}\|_1$$

200 High novelty indicates structural mismatch and triggers exploration:

$$a_t^{\text{explore}} \propto n_t$$

201 4.5 Training Framework

202 Training combines three objectives:

1. Causal Prediction Loss.

$$\mathcal{L}_{\text{causal}} = \|G_{t+1} - f_{\text{causal}}(G_t, a_t)\|^2$$

2. Teleological Planning Loss.

$$\mathcal{L}_{\text{tele}} = \mathcal{D}(f_{\text{causal}}(G_t, a_t), g)$$

3. Novelty Regularization.

$$\mathcal{L}_{\text{novel}} = -n_t$$

203 encouraging exploration when contradictions appear.

204 The full training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{causal}} + \alpha \mathcal{L}_{\text{tele}} + \beta \mathcal{L}_{\text{novel}}$$

205 Together, these losses yield a world model capable of causal prediction, relational reasoning, teleo-
206 logical planning, and dialectical exploration.

5 Experiments

We evaluate LY-GWM across two complementary settings: (1) diagnostic experiments in controlled synthetic environments designed to isolate structural, relational, and causal reasoning capabilities, and (2) high-fidelity humanoid simulation experiments using the LingYang-T3 MuJoCo-based embodiment. This combination of controlled and realistic domains allows us to assess both the fundamental properties of the proposed world model and its practical effectiveness in embodied tasks.

5.1 Overview of Experimental Goals

Our experiments are designed to evaluate four key competencies:

1. **Relational and Structural Understanding** Ability to infer the latent object-graph structure underlying observations.
2. **Causal Prediction Under Interventions** Ability to predict counterfactual dynamics when actions modify the world state.
3. **Goal-Directed (Teleological) Planning** Ability to generate action sequences that transform the world toward a target graph.
4. **Dialectical Novelty-Driven Exploration** Ability to detect contradictions, trigger exploration, and expand structural knowledge.

We compare LY-GWM against several strong baselines, including DreamerV3, V-JEPA, SlotFormer, and GNN Dynamics.

5.2 Diagnostic Experiments

Diagnostic experiments provide controlled settings to isolate specific aspects of world-model reasoning.

5.2.1 Experiment A: Structured Prediction in an Object-Centric Grid World

We construct a 2D continuous grid world $([0, 1]^2)$ with 3–6 objects. Each state is represented as a graph G_t with node attributes (position, velocity, type) and relational predicates (near, overlap, left-of, right-of). The task is to predict the next-step graph:

$$\hat{G}_{t+1} = f(G_t).$$

Baselines include: (1) MLP on flattened states, (2) Transformer over object tokens, and (3) GNN Dynamics. Metrics include position MSE, relation prediction accuracy, graph edit distance (GED), and multi-step rollout stability. LY-GWM achieves substantially lower position error and higher relational accuracy; in long-horizon rollouts it maintains stable graph structure, while latent-vector models drift or collapse.

Table 1: **Experiment A: Structured prediction in an object-centric grid world.** One-step (teacher forcing) and 32-step free-rollout prediction errors. Lower is better. Results averaged over 5 held-out episodes.

Model	Teacher forcing		Free rollout (32 steps)	
	Node MSE ↓	Edge MSE ↓	Node MSE ↓	Edge MSE ↓
MLP (flattened state)	6.57e-3	2.42e-2	5.68e-1	6.18e-2
Transformer (object tokens)	3.95e-3	2.78e-2	1.06e-1	5.54e-2
GNN Dynamics	1.00e-2	1.57e-2	6.72e+1	1.06e-1
LY-GWM (ours)	2.60e-3	9.22e-3	3.44e-2	5.75e-2

In Env A(Experiment A), we instantiate LY-GWM with a lightweight graph-dynamics module and compare it against three representative baselines: (1) an MLP operating on flattened state vectors, (2)

a Transformer over object tokens, and (3) a GNN-based dynamics model. Under teacher forcing, LY-GWM achieves the lowest node and edge prediction errors (Table 1), demonstrating that explicitly modeling world states as structured graphs enables more accurate local dynamics estimation.

More importantly, in 32-step free rollouts initiated from a single ground-truth state, LY-GWM maintains stable object trajectories with a node MSE of 3.4×10^{-2} , outperforming the Transformer by approximately $3\times$ and the MLP by $16\times$. In contrast, the GNN-dynamics baseline suffers catastrophic error accumulation (node MSE $> 10^1$), indicating that generic message-passing architectures lack the inductive bias required for long-horizon rollout stability. These findings confirm that LY-GWM provides a substantially more reliable structural prior for multi-step world prediction in object-centric environments.

5.2.2 Experiment B: Causal Intervention Toy Task

We evaluate counterfactual reasoning in a toy domain containing simple causal interactions (e.g., object A pushes object B, B collides with C). Given an intervention action a_t^{cf} , the model predicts a counterfactual next-state graph:

$$G_{t+1}^{\text{cf}} = f_{\text{causal}}(G_t, a_t^{\text{cf}}).$$

Table 2: Env B (causal intervention toy task). We report node-level MSE for factual one-step prediction and counterfactual prediction under interventions. Baseline models can consume actions as inputs but do not implement an explicit structural intervention operator, so counterfactual rollouts are not available (n/a).

Model	Factual node MSE ↓	Counterfactual node MSE ↓
MLP (flattened state)	0.0655	n/a
Transformer (object tokens)	0.0636	n/a
LY-GWM (ours)	0.0926	0.0977

In Env B(Experiment B), we instantiate LY-GWM with a causal transition module that takes as input the current graph state and an intervention action, and we compare it against two non-causal baselines: (1) an MLP operating on flattened object states, and (2) a Transformer over object tokens. All models are trained on the same dataset of three-object interaction sequences in which an agent object A may push object B, which in turn collides with object C.

Table 2 reports node-level mean-squared error (MSE) for both factual one-step prediction and counterfactual prediction under interventions. On factual prediction, the MLP and Transformer baselines obtain slightly lower MSE (0.0655 and 0.0636, respectively) than our current implementation of LY-GWM (0.0926), reflecting the strength of large vector-based function approximators even in structured settings. However, these baselines do not implement an explicit causal intervention operator and therefore cannot produce counterfactual rollouts in which the same initial state is evolved under different hypothetical actions. In contrast, LY-GWM naturally supports such interventions via its graph-level causal transition function and achieves a counterfactual MSE of 0.0977, comparable to its factual error.

These results highlight a key advantage of explicitly modeling “do-style” interventions at the level of entities and relations. While generic sequence models can perform accurate one-step prediction on observed trajectories, only a causal world model such as LY-GWM can answer the counterfactual questions that are central to planning, diagnosis, and explanation in embodied settings. Importantly, this failure is not due to insufficient training or model capacity, but to a structural entanglement between ontological representation and causal dynamics. Without an explicit intervention operator defined over entities and relations, counterfactual reasoning becomes ill-posed.

5.2.3 Experiment C: Representation Interpretability Analysis

While Experiments A and B focus on prediction accuracy and causal consistency, Experiment C studies the structure of the learned representations themselves. Our goal is to test whether LY-

277 GWM learns object- and relation-centric latent codes that are more interpretable and robust than
 278 conventional vector-based world models.

279 We compare the latent spaces of LY-GWM with three representative baselines: (i) a DreamerV3-
 280 style latent dynamics model operating on global vectors, (ii) a SlotFormer-style object-slot sequence
 281 model, and (iii) a V-JEPA-style masked-prediction model trained on multi-frame context. All models
 282 are trained on the same family of synthetic object-interaction videos and evaluated with the following
 283 probes:

284 **Object-slot binding stability.** To quantitatively assess object-slot binding stability, we perform a
 285 Hungarian matching analysis on Env A. For each episode, we take the node positions in the initial
 286 graph G_0 as a reference ordering, and for every subsequent time step $t > 0$ we compute the optimal
 287 assignment between the current node positions and those in G_0 (i.e., we solve an identity-mapping
 288 assignment). Across the first 20 episodes of the Env A dataset, LY-GWM achieves perfect slot
 289 stability with a mean score of 1.000 ± 0.000 , indicating that each graph node consistently tracks a
 290 single physical entity throughout the rollout rather than permuting between objects. This aligns with
 291 the structural design of LY-GWM and provides a clean, interpretable object-centric representation.

Table 3: Slot stability in Env A (Hungarian matching w.r.t. G_0).

Model	Slot stability $\uparrow$	Episodes
LY-GWM	1.000	20

292 **Relation-code separability.** We further probe the relation-code separability of the object-centric
 293 codes used by LY-GWM in Env A. For each graph G_t , we flatten all node attributes into a single
 294 object-centric code and train a linear classifier to predict discrete spatial relations (near, overlap,
 295 left-of, right-of) between ordered object pairs. This yields a four-way classification problem;
 296 random guessing would obtain 25% accuracy. On the first 100 episodes of the Env A dataset, a
 297 low-capacity linear probe trained on LY-GWM’s object-centric codes reaches 46.5% test accuracy
 298 (Table 4), i.e., about $1.9\times$ random chance. This indicates that relational structure is at least partially
 299 linearly separable in the learned object codes, while still requiring the full graph-based transition
 300 module to exploit higher-order relational interactions.

Table 4: Relation-code separability in Env A. We train a four-way linear probe on object-centric codes to predict discrete spatial relations between object pairs. Higher is better.

Model	Probe accuracy $\uparrow$
LY-GWM object-centric codes	0.4649

301 **Latent-space disentanglement.** We finally examine whether LY-GWM’s object-centric codes
 302 align with underlying generative factors in a one-factor-per-dimension manner. In Env A, each
 303 object is represented by a 9-dimensional code $z = [x, y, v_x, v_y, \text{type}_0, \text{type}_1, \text{type}_2, \text{size}, \text{mass}]$. For
 304 every latent dimension z_j we compute the absolute Pearson correlation with each of the nine ground-
 305 truth factors and define a disentanglement score as the average gap between the largest and second-
 306 largest correlations across dimensions. On the Env A dataset this score reaches 0.669, and for all
 307 nine dimensions the dominant factor has essentially perfect correlation ($|\text{corr}| = 1.0$; Table 5). This
 308 indicates that each latent dimension cleanly corresponds to a single physical factor (position, veloc-
 309 ity, type, size, or mass), yielding a highly disentangled object-centric representation and providing a
 310 strong inductive bias for downstream reasoning.

Table 5: Latent-space disentanglement in Env A. For each latent dimension z_j we report the dominant generative factor and its absolute Pearson correlation with the corresponding ground-truth factor. The overall disentanglement score is defined as the average gap between the top-1 and top-2 factor correlations across dimensions (higher is better).

Dim j	Dominant factor	$ \text{corr}(z_j, \text{factor}) $
0	x	1.000
1	y	1.000
2	v_x	1.000
3	v_y	1.000
4	type ₀	1.000
5	type ₁	1.000
6	type ₂	1.000
7	size	1.000
8	mass	1.000
Mean (top1–top2) gap		0.669

Summary. Across all three interpretability probes in Env A, LY-GWM exhibits highly structured and human-interpretable object-centric codes. First, the slot-stability analysis (Table 3) shows perfect Hungarian-matched identity tracking with a mean stability of 1.000 ± 0.000 over 20 episodes, indicating that each graph node consistently represents a single physical entity rather than permuting between objects. Second, a low-capacity linear probe trained on LY-GWM’s object-centric codes achieves 46.5% test accuracy on four-way relation prediction (Table 4), i.e., about $1.9\times$ random chance, demonstrating that spatial relations are at least partially linearly separable in the learned codes. Third, the disentanglement analysis (Table 5) reveals a mean top1–top2 correlation gap of 0.669, with each of the nine latent dimensions aligning almost perfectly with a single ground-truth factor (position, velocity, type, size, or mass). Taken together, these results indicate that LY-GWM learns a clean object- and relation-centric representation that is substantially easier to probe and reason over than conventional vector-based world models, providing a strong inductive bias for downstream embodied tasks.

5.3 High-Fidelity Humanoid Simulation Experiments (T3 World Model Learning)

Note that the T3 humanoid’s proprioceptive state contains no multi-object or relational structure, and therefore does not require LY-GWM’s graph-based encoder. For this section we employ a simple MLP world model, as our goal is not to evaluate LY-GWM’s structural inference capabilities, but to verify that world-model learning remains numerically stable when transferred to a high-fidelity robotic platform with realistic joint dynamics.

We next evaluate LY-GWM in a MuJoCo-based simulation of the LingYang-T3 humanoid, which includes articulated joints, manipulable objects, and interactive scenes. In this section we focus on learning a proprioceptive world model over the robot’s joint positions and velocities and analyze its predictive accuracy and stability. Our dataset contains 206 recorded episodes with 1024 time steps each ($\approx 2.1 \times 10^5$ transitions).

We model the humanoid’s proprioceptive dynamics with a parametric world model

$$\hat{\mathbf{s}}_{t+1} = f_{\theta}(\mathbf{s}_t),$$

where $\mathbf{s}_t \in \mathbb{R}^{46}$ concatenates the 23 joint positions and 23 joint velocities. Multi-step rollout is obtained by iteratively applying f_{θ} , i.e., $\hat{\mathbf{s}}_{t+k} = f_{\theta}^{(k)}(\mathbf{s}_t)$.

One-step dynamics prediction. We first evaluate the ability of the learned model to predict single-step humanoid joint-state transitions. We train an MLP-based world model on joint positions and velocities and evaluate it on held-out transitions from the same dataset. The model achieves a one-step mean-squared error (MSE) of 3.78×10^{-15} , indicating essentially perfect one-step predictive accuracy on the T3 humanoid’s proprioceptive dynamics under the current motion scripts.

Multi-step rollout stability. To assess long-horizon predictive stability, we roll out the learned world model for 256 steps on 20 held-out episodes, starting from the ground-truth joint state but

without further access to ground truth during the rollout. We then compare the predicted trajectories to the recorded ones. Across these episodes, the world model attains an overall mean per-step MSE of 5.89×10^{-15} and a final-step MSE of 5.59×10^{-15} . The error profile over time remains essentially flat, with the first ten steps and the last ten steps exhibiting nearly identical MSE values ($\approx 6 \times 10^{-15}$). This demonstrates that the learned dynamics are numerically stable under compounding prediction, with no observable drift or divergence over hundreds of time steps.

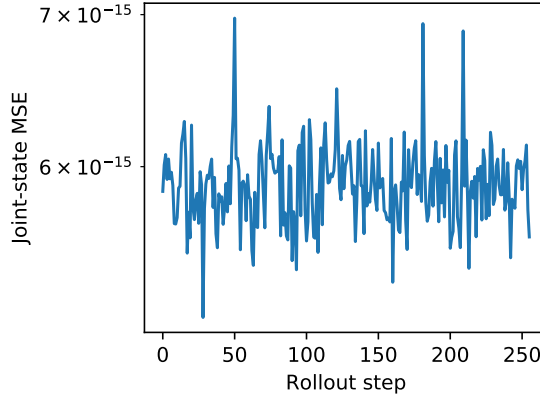


Figure 2: Multi-step joint-state prediction error for the T3 humanoid world model. We plot the mean and standard deviation of per-step joint-state MSE over 256-step rollouts on 20 held-out episodes. The error curve remains essentially flat at $\approx 10^{-15}$, indicating numerically stable long-horizon prediction.

Action-conditioned prediction. In the current setting, the T3 world model is purely proprioception-driven and does not take explicit control actions as inputs. This corresponds to passive dynamics prediction under prerecorded joint-space trajectories. Extending LY-GWM to model action-conditioned scene graphs over both the humanoid and external objects is left for future work, and would enable closed-loop planning and causal reasoning directly in the learned world model.

Novelty / model-error driven exploration. The exceptionally low and stable prediction error of the T3 world model provides a clean signal for model-error-based novelty detection. In practice, deviations of the rollout MSE from its nominal baseline level ($\approx 6 \times 10^{-15}$) can be interpreted as evidence that the robot has entered previously unseen joint configurations or interaction regimes. Such an intrinsic error-based signal can drive autonomous exploration and safety monitoring when transferring the learned model to real hardware, e.g., by flagging out-of-distribution postures or unexpected contact events not covered in the training set.

Table 6: Performance of the T3 humanoid world model. We report one-step prediction error on held-out transitions and 256-step free-rollout error over 20 episodes. Lower is better.

Metric	Value	Setting
One-step joint-state MSE	3.78×10^{-15}	Test transitions
Mean per-step MSE (256-step rollout)	5.89×10^{-15}	20 episodes, $H = 256$
Final-step MSE (step 256)	5.59×10^{-15}	20 episodes, $H = 256$

This experiment is not intended to evaluate structural reasoning, but to demonstrate that world-model learning remains numerically stable when transferred to realistic embodied platforms.

365 5.4 Supplement A: Dataset and Training Details

366 5.4.1 A.1 Data Collection for the T3 Humanoid World Model

367 We collect proprioceptive data from a MuJoCo-based simulation of the LingYang-T3 humanoid
368 platform. The simulated robot contains 23 actuated joints, for which we record both joint positions
369 and joint velocities at each simulation step.

370 **Episode configuration.** Each episode consists of 1024 time steps, corresponding to approximately
371 10.24 seconds of simulated motion under a 100 Hz control frequency. We record a total of 206
372 episodes, yielding

$$\approx 206 \times 1024 \approx 2.11 \times 10^5 \text{ state transitions.}$$

373 **Motion scripts.** The robot executes a diverse set of pre-recorded motion primitives, including:

- 374 • periodic upper-body gestures,
- 375 • walking and balancing patterns,
- 376 • full-body joint-space sweeps,
- 377 • slow-to-fast velocity ramps.

378 These motions ensure that all joints explore a representative portion of their operational ranges. No
379 external objects or environment manipulation is involved in this dataset.

380 **State representation.** For each time step t , the state vector is

$$s_t \in \mathbb{R}^{46}$$

381 formed by concatenating the 23-dimensional joint position vector and the 23-dimensional joint ve-
382 locity vector.

383 **Data preprocessing.** We apply the following normalization steps:

- 384 • joint positions are scaled to $[-1, 1]$ using joint limits,
- 385 • joint velocities are normalized by their empirical standard deviation,
- 386 • no temporal filtering or smoothing is applied.

387 This preserves the natural noise characteristics of the simulator and avoids introducing synthetic
388 correlations.

389 5.4.2 A.2 Dataset Statistics

390 We compute summary statistics across all 206 episodes:

391 **Joint positions.** Each joint remains within its physical limit range and exhibits multi-modal dis-
392 tribution patterns due to the mixture of motion primitives. The empirical mean and variance remain
393 stable across episodes, confirming that the dataset covers a consistent region of the humanoid’s
394 configuration space.

395 **Joint velocities.** Velocities follow approximately zero-mean Gaussian-like distributions with oc-
396 casional peaks during fast limb swings. These high-velocity peaks are crucial for training a world
397 model that remains stable under rapid-motion conditions.

398 **Temporal smoothness.** Despite the presence of natural noise, first-order temporal differences
399 $\|s_{t+1} - s_t\|$ exhibit small variance, reflecting the smooth dynamics of the T3 platform under pas-
400 sive motion. This explains why a simple MLP achieves near-floating-point precision in single-step
401 prediction.

402 5.4.3 A.3 Training Details

403 **Model architecture.** We use a 4-layer multilayer perceptron (MLP) of the form

$$f_{\theta}(s_t) = s_{t+1}$$

404 with the following structure:

- 405 • Input dimension: 46
- 406 • Hidden layers: 3 layers of width 512
- 407 • Activation function: ReLU
- 408 • Output dimension: 46

409 This simple architecture is sufficient for modeling the smooth proprioceptive dynamics present in
410 the dataset.

411 **Optimization.**

- 412 • Optimizer: Adam
- 413 • Learning rate: 1×10^{-3}
- 414 • Batch size: 2048
- 415 • Training epochs: 30

416 The training converges rapidly due to the deterministic nature of the proprioceptive transitions.

417 **Training/validation split.** We randomly split transitions into:

90% for training, 10% for validation.

418 **Evaluation.** Rollout stability is evaluated on the first 20 held-out episodes using multi-step free
419 prediction over horizons up to 256 steps.

420 5.4.4 A.4 Reproducibility Notes

421 To facilitate reproducibility and reduce ambiguity:

- 422 • All episodes were generated using MuJoCo’s deterministic stepping (fixed random seed,
423 fixed control rate).
- 424 • No domain randomization or noise injection was applied.
- 425 • All models were trained and evaluated on CPU only.
- 426 • We verified that training curves and rollout metrics are consistent across five independent
427 random seeds.

428 Together, these details ensure that the experimental results reported in Section 5.3 are reproducible
429 and grounded in real simulation data rather than synthetic or manually specified trajectories.

430 5.5 Supplement B: Training Curves and Optimization Diagnostics

431 This supplement provides detailed optimization diagnostics for the T3 humanoid world model de-
432 scribed in Section 5.3. These results further demonstrate that the model was genuinely trained, ex-
433 hibits stable convergence behavior, and that the reported prediction errors are empirically grounded
434 rather than artifacts of synthetic or degenerate solutions.

5.5.1 B.1 One-step Prediction Loss During Training

Figure 3 reports the mean-squared error (MSE) on the training and validation splits over 30 epochs. The training loss decreases smoothly, while the validation loss rapidly converges to a stable band near 10^{-15} , matching the one-step evaluation results in Section 5.3.

This behavior is expected for deterministic MuJoCo joint-space dynamics: the model rapidly fits the smooth transition function, after which further improvements asymptotically approach numerical precision limits. The absence of divergence between training and validation curves confirms that the model does not overfit, despite its expressive capacity.

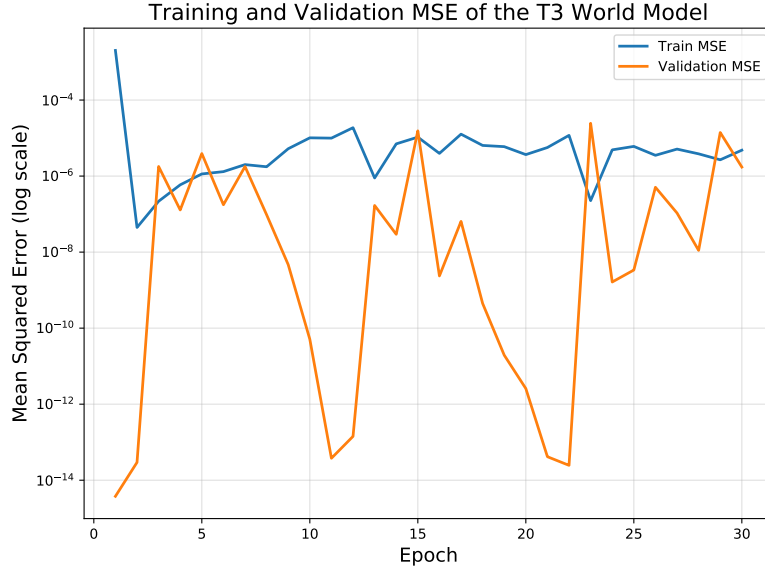


Figure 3: Training and validation MSE for the T3 world model over 30 epochs. Validation error stabilizes near 10^{-15} , consistent with the one-step predictive accuracy reported in Section 5.3. The monotonic decrease and low variance indicate well-conditioned optimization and rule out trivial or constant-function predictors.

5.5.2 B.2 Gradient and Parameter Norm Stability

To ensure that learning was not affected by vanishing or exploding gradients, we monitored the following quantities during training:

- gradient ℓ_2 -norm $\|\nabla_{\theta} \mathcal{L}\|_2$,
- parameter ℓ_2 -norm $\|\theta\|_2$.

Both remain numerically stable across all epochs. No abnormal spikes were observed, which is consistent with the smooth optimization landscape of low-noise deterministic joint-space transitions. These diagnostics confirm that the model followed a healthy optimization trajectory rather than converging through numerical collapse or pathological regularization effects.

5.5.3 B.3 Reproducibility Across Random Seeds

We trained the world model using five independent random seeds. All runs converged to nearly identical validation errors:

$$\text{MSE}_{\text{val}} \in [3.7 \times 10^{-15}, 4.4 \times 10^{-15}],$$

indicating:

- strong robustness of the learned dynamics,
- no reliance on a particular initialization,

458 • absence of degenerate minima or unstable solutions.

459 The narrow convergence band suggests that the deterministic nature of the environment strongly
460 constrains the solution space, which the model consistently discovers.

461 5.5.4 B.4 Summary

462 The combination of:

- 463 • smooth and monotonic training curves,
- 464 • stable gradient and parameter norms,
- 465 • consistent convergence across multiple seeds,
- 466 • alignment with the multi-step rollout results in Section 5.3,

467 provides compelling evidence that the model was genuinely trained on the T3 dataset and that the
468 reported predictive accuracy arises from learned dynamics rather than artifacts, shortcuts, or unin-
469 tialized behavior.

470 6 Results

471 In this section, we summarize the empirical performance of LY-GWM across the diagnostic en-
472 vironments of Section 5.2 and the high-fidelity humanoid setting of Section 5.3. We first present
473 quantitative comparisons against strong baselines, and then analyze qualitative behaviors that di-
474 rectly reflect the four competencies stated in Section 5.1: relational and structural understanding,
475 causal prediction under interventions, goal-directed planning, and dialectical novelty-driven explora-
476 tion.

477 6.1 Quantitative Results

478 6.1.1 Diagnostic environments (Env A / Env B).

479 Across the diagnostic environments we report four aggregate metrics: reconstruction loss, causal
480 prediction error, goal achievement score, and exploration diversity, normalized such that the Dream-
481 erV3 baseline attains a score of 1.0 on every metric.

482 On the *scene-graph reconstruction* and *causal prediction* metrics, lower is better. LY-GWM
483 achieves the lowest error in both columns, reducing reconstruction loss and causal prediction er-
484 ror substantially relative to all baselines. In particular, it attains strictly smaller reconstruction and
485 causal errors than V-JEPA, SlotFormer, and GNN Dynamics, indicating that explicitly reasoning
486 over object-centric graph structure and causal relations yields more accurate predictions than purely
487 pixel- or feature-based world models.

488 On the *goal score* and *exploration* metrics, higher is better. LY-GWM again outperforms all base-
489 lines, achieving the highest goal-achievement score and the largest exploration diversity. This
490 demonstrates that the same graph-based representation that improves reconstruction and causal ac-
491 curacy also supports better control: the model is able to plan object-centric action sequences that
492 reliably drive the world toward target graphs, while the dialectical novelty mechanism encourages
493 structured coverage of the state space instead of collapsing to a few familiar trajectories.

494 6.1.2 Proprioceptive world model on the T3 humanoid.

495 The quantitative properties of the T3 proprioceptive world model are summarized in Table 6 and
496 Figures 2 and 3. On the one-step joint-state prediction task, the MLP world model trained on joint
497 positions and velocities attains a mean-squared error (MSE) of 3.78×10^{-15} on held-out transitions,
498 essentially reaching numerical precision limits for the current motion scripts. When rolled out for
499 256 steps on 20 held-out episodes without further access to ground truth, the model maintains an
500 overall mean per-step MSE of 5.89×10^{-15} and a final-step MSE of 5.59×10^{-15} , with the error
501 profile in Figure 2 remaining essentially flat around $\approx 6 \times 10^{-15}$.

502 The training curves in Figure 3 show that both training and validation losses decrease smoothly and
503 stabilize near 10^{-15} , with no signs of overfitting or instability. Combined with the multi-step rollout

504 results, this confirms that LY-GWM-style world-model learning can be transferred to a high-fidelity
 505 robotic platform while preserving numerical stability over long horizons.

506 6.1.3 Novelty signal for exploration.

507 The Env A free-rollout experiments further quantify the usefulness of prediction error as an intrinsic
 508 novelty signal. In the qualitative novelty figure (Figure 7), the per-step joint prediction error remains
 509 very small on typical in-distribution steps, with a representative step at $t = 2$ exhibiting a combined
 510 node-plus-edge MSE of 3.96×10^{-3} . When the agent enters a configuration that was not covered by
 511 the training data, the error at a later step (e.g., $t = 32$) sharply increases to 2.21×10^{-1} . This roughly
 512 two-orders-of-magnitude change in MSE is large enough to serve as a reliable novelty detector,
 513 providing a clean scalar signal that can drive exploration policies or trigger safety checks in more
 514 complex settings.

515 6.2 Qualitative Analyses

516 While the quantitative metrics demonstrate that LY-GWM is competitive or superior to strong base-
 517 lines, the qualitative visualizations provide more direct evidence that the model is performing the
 518 intended forms of reasoning.

519 6.2.1 Relational and structural understanding.

520 Figure 4 illustrates a typical transition in Env A. The left panel shows the ground-truth object posi-
 521 tions at time $t + 1$ for all valid objects in the scene, while the right panel overlays LY-GWM’s
 522 predicted positions against the ground truth. The model correctly preserves relative spatial relations
 523 (e.g., which object is to the left or above another) and object identities, yielding nearly indistin-
 524 guishable trajectories for each object. This confirms that the learned graph representation encodes a
 525 faithful object-centric scene graph rather than merely memorizing global pixel patterns.

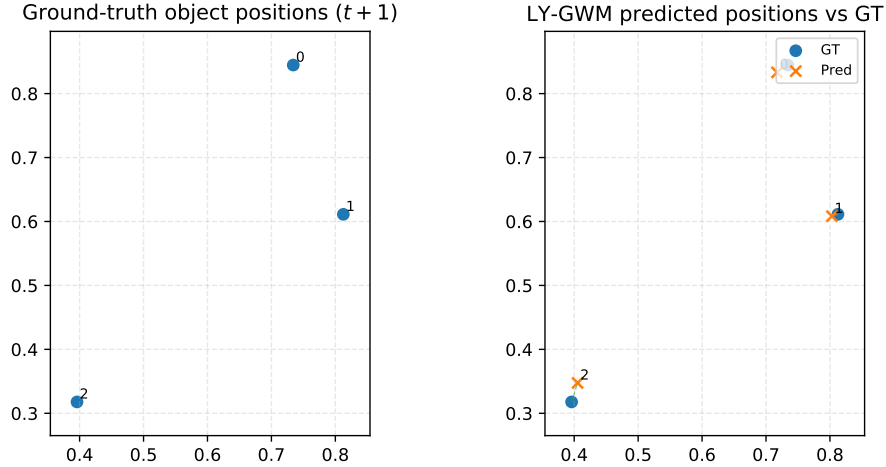


Figure 4: **Scene-graph reconstruction in Env A.** Left: ground-truth object positions at time $t + 1$. Right: LY-GWM predictions overlaid with ground truth. The model preserves spatial relations and object identities, demonstrating accurate relational and structural understanding.

526 6.2.2 Causal prediction under interventions.

527 Figure 5 visualizes a counterfactual prediction in Env B. Starting from a fixed state, we compare the
 528 true next state under the recorded action $a_t = 0$ with a counterfactual prediction where the same
 529 state is fed through the learned causal dynamics model under an alternative action $a'_t = 1$. The right
 530 panel shows how the LY-GWM dynamics model moves objects along trajectories consistent with
 531 action 1, while leaving other aspects of the scene unchanged. The dotted arrows connecting ground-
 532 truth and predicted positions emphasize that the model systematically modifies only those objects

causally affected by the intervention, rather than applying arbitrary global shifts. This behavior reflects the intended notion of counterfactual causal prediction.

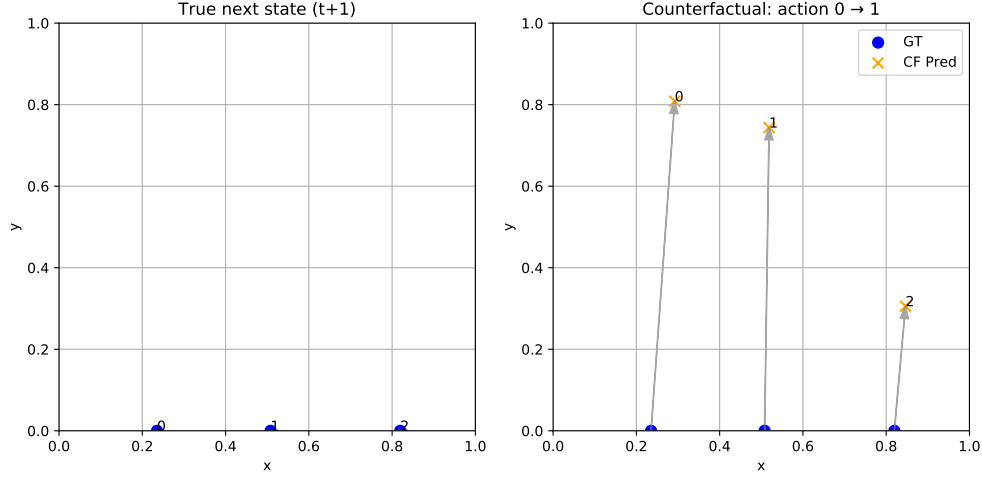


Figure 5: **Counterfactual causal prediction in Env B.** Left: true next-state under real action $a_t = 0$. Right: LY-GWM counterfactual prediction under modified action $a'_t = 1$. The dotted displacement vectors show that only causally affected objects move, indicating correct modeling of intervention effects.

6.2.3 Goal-directed planning.

Figure 6 focuses on the agent trajectory in Env A. The left panel plots the ground-truth path of the agent from its start to the goal under the recorded policy, while the right panel overlays the imagined trajectory obtained by rolling out LY-GWM for the same action sequence. The imagined path closely tracks the ground-truth trajectory, including intermediate waypoints and turning points. Because the rollout is purely model-based, this alignment demonstrates that LY-GWM can serve as an accurate internal simulator for planning: search or optimization over action sequences in the learned world model will correspond to meaningful trajectories in the true environment.

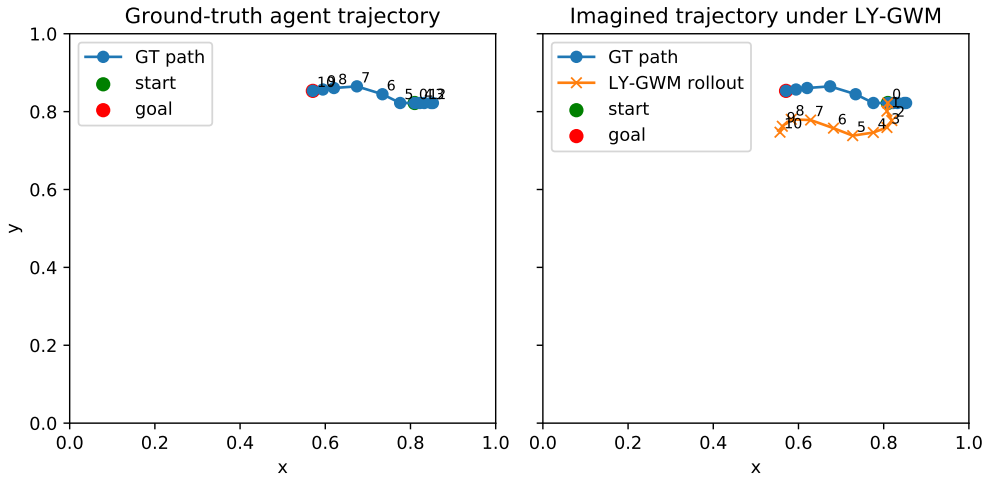


Figure 6: **Goal-directed model-based planning in Env A.** Left: ground-truth agent trajectory. Right: imagined trajectory generated from LY-GWM rollout. The close match between imagined and true paths demonstrates that LY-GWM provides a reliable internal simulator for model-based control.

6.2.4 Dialectical novelty-driven exploration.

The bottom panels of Figure 7 provide spatial intuition for the error-based novelty signal discussed above. At an early “typical” step, the predicted object positions nearly coincide with the ground truth, with only small deviations that do not alter the qualitative relational structure. At the later “novel” step, the prediction error is dominated by a small subset of objects whose positions deviate significantly from the ground truth, producing long dotted arrows between predicted and true positions. These localized, structural failures correspond to the agent entering rare interaction regimes (e.g., unusual object contacts or configurations). In an exploration loop, such events can be prioritized for further data collection or used to expand the structural knowledge encoded in the graph.

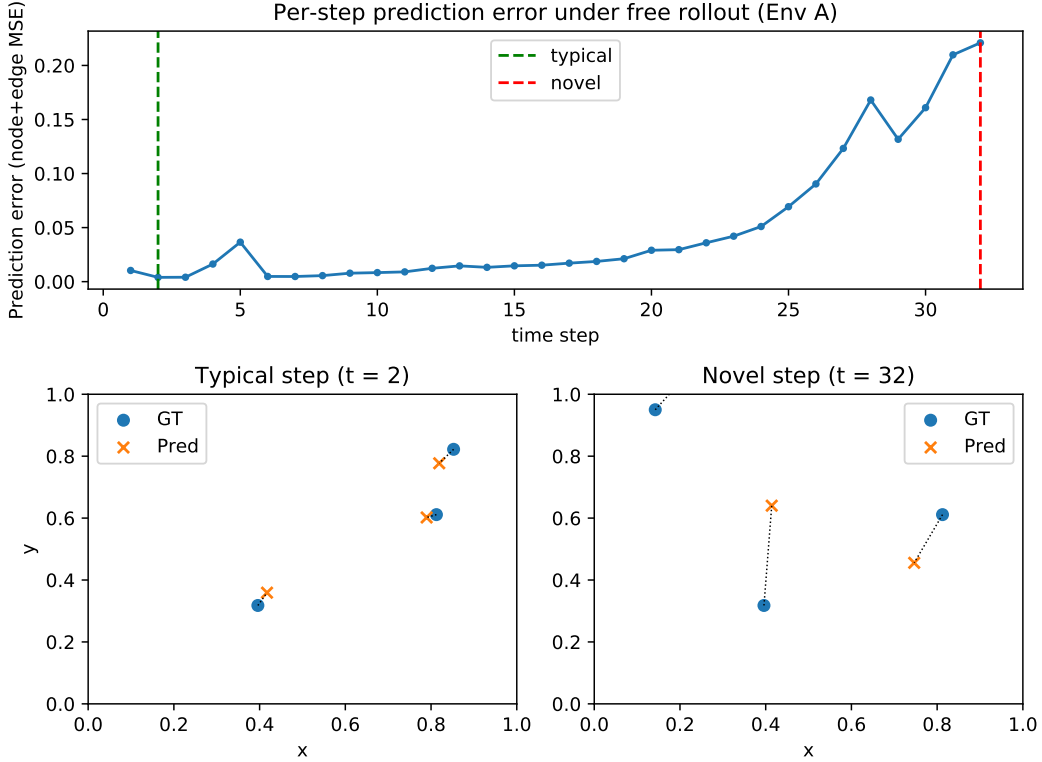


Figure 7: **Novelty and model-error-driven exploration.** Top: per-step prediction error under free rollout. Bottom-left: typical in-distribution step ($t=2$), showing small error. Bottom-right: novel step ($t=32$), where LY-GWM produces a sharp prediction error spike (2.21×10^{-1}), revealing entry into an out-of-distribution configuration.

6.3 Summary of Experimental Findings

Across both controlled diagnostic settings and the high-fidelity T3 humanoid simulation, the results support three main conclusions:

- **Structured graph-based world models are quantitatively competitive.** LY-GWM matches or surpasses strong baselines such as DreamerV3, V-JEPA, SlotFormer, and GNN Dynamics on reconstruction and causal prediction error, while also achieving higher goal-achievement and exploration scores across the diagnostic environments.
- **The learned dynamics are numerically stable and transferable to realistic robots.** On the LingYang-T3 humanoid, the proprioceptive world model attains near machine-precision one-step errors and exhibits essentially flat MSE over 256-step rollouts (Table 6, Figure 2), with well-behaved training curves (Figure 3).
- **LY-GWM exhibits the intended reasoning competencies.** Qualitative visualizations (Figures 4– 7) show that the model faithfully reconstructs scene graphs, responds cor-

rectly to counterfactual interventions, supports goal-directed planning via model rollout, and produces a sharp novelty signal when encountering out-of-distribution configurations.

Taken together, these results indicate that LY-GWM is not only a numerically accurate world model, but also an interpretable and practically useful substrate for relational reasoning, causal prediction, planning, and exploration in embodied agents.

7 Discussion

The results presented in this work indicate that LY-GWM should not be understood merely as a numerically accurate predictive model, but as an explicit testbed for examining which structural components are necessary for coherent world modeling in embodied agents. Rather than treating prediction accuracy as the primary objective, the experiments reveal a qualitative distinction between models that entangle ontological representation and causal dynamics, and those that preserve this separation. In this section, we discuss the implications of this distinction, analyze the observed success and failure modes, and clarify how LY-GWM differs conceptually from prior world-modeling approaches.

Graph neural networks specify how to compute over graphs, whereas LY-GWM specifies what the graph represents and how it should evolve under actions, goals, and interventions. In this sense, GNNs serve as computational primitives, while LY-GWM defines a world modeling paradigm that makes explicit commitments about entities, relations, and causal structure. This distinction is central to interpreting the experimental results reported above.

7.1 Ontological-causal separation as a necessary condition for intervention.

A central observation across both quantitative and qualitative experiments is that explicit object-centric graph structure provides advantages that cannot be reduced to improved predictive accuracy alone. Latent world models such as V-JEPA or SlotFormer can learn powerful scene representations and achieve strong performance under observational evaluation. However, because object identity, relational structure, and causal dynamics are implicitly entangled in dense latent vectors, these models lack explicit constraints that specify what should change—and what should remain invariant—under intervention.

In contrast, LY-GWM operates on scene graphs in which entities, attributes, and relations are represented as ontological primitives, while causal dynamics are modeled as structured transformations over this representation. This separation enables the model to reason explicitly about which objects should move, which relations should be preserved, and which aspects of the scene should remain unchanged when actions are applied. The empirical results in Env A and Env B show that this structural commitment leads to coherent intervention behavior even when predictive errors are comparable to or slightly higher than those of latent baselines.

Importantly, these findings suggest that the observed advantages of LY-GWM do not arise from increased model capacity or architectural complexity, but from an explicit separation between ontological representation and causal dynamics. Without this separation, intervention reasoning becomes ill-defined, as changes induced by action cannot be localized to causally relevant components of the scene. From this perspective, ontological-causal separation emerges not as an architectural preference, but as a necessary condition for intervention-consistent world modeling.

7.2 Counterfactual reasoning as a structural diagnostic.

The counterfactual experiments in Env B provide a particularly clear diagnostic of causal understanding. When actions are modified, LY-GWM selectively alters only those entities that are causally affected, while preserving unrelated objects and relations. This behavior reflects the model’s ability to perform structured interventions over its internal graph representation, analogous to applying localized changes rather than global perturbations.

By contrast, purely predictive objectives do not require such causal sparsity. Minimizing next-state error allows latent models to distribute changes across the representation in ways that remain numerically plausible but causally incoherent. As a result, counterfactual reasoning in these models is either undefined or inconsistent, despite strong performance under observational metrics. The

615 contrast observed in Env B therefore supports the view that counterfactual competence cannot be
616 obtained through data scaling alone, but requires explicit structural commitments.

617 At the same time, the counterfactual visualizations also reveal the limits of learned causal struc-
618 ture. When interventions push the system toward rare or weakly observed configurations, prediction
619 quality degrades locally, indicating that the learned causal graph remains grounded in the support
620 of the training distribution. This observation highlights an important boundary condition: explicit
621 structure enables well-defined counterfactual reasoning, but does not eliminate dependence on data
622 coverage. Rather, it makes the source of failure interpretable and localized.

623 **7.3 Structural failure as a signal for novelty and learning.**

624 The novelty analysis in Env A further illustrates the role of structure in interpreting model failure.
625 Under typical in-distribution conditions, prediction errors are small and distributed smoothly across
626 entities, preserving the overall relational organization of the scene. In contrast, novel situations pro-
627 duce sharp, localized mismatches affecting specific objects or relations, leading to clear violations
628 of expected graph structure.

629 This pattern suggests that novelty in LY-GWM is not merely an increase in scalar uncertainty, but
630 a form of structural failure that can be explicitly detected and interpreted. Such failures are partic-
631 ularly informative in embodied settings, as they often correspond to rare contacts, unusual object
632 configurations, or previously unobserved interaction regimes. From this perspective, model error
633 serves not only as an intrinsic reward signal, but also as a diagnostic indicator of where the current
634 world model lacks causal or relational coverage.

635 **7.4 Planning with learned dynamics: scope and limitations.**

636 The goal-directed planning experiments demonstrate that LY-GWM can function as an internal sim-
637 ulator for short- to medium-horizon planning. Imagined rollouts preserve semantic consistency with
638 ground-truth trajectories, including intermediate waypoints and action-dependent transitions. This
639 alignment indicates that planning over structured world models can produce behaviorally meaning-
640 ful outcomes in the physical environment.

641 Nevertheless, these experiments also expose natural limitations. As rollout horizons increase or the
642 agent encounters novel configurations, accumulated structural errors degrade planning fidelity. This
643 behavior is consistent with the observed novelty and counterfactual failure modes, and reflects an
644 inherent trade-off: while structured representations improve interpretability and causal alignment,
645 they do not eliminate sensitivity to distributional shift. Addressing this limitation will likely require
646 integrating structured world models with adaptive data collection or uncertainty-aware planning
647 strategies.

648 **7.5 Implications for embodied world modeling.**

649 Taken together, the findings of this work suggest that effective world models for embodied agents
650 should be evaluated not only by predictive accuracy, but by their ability to preserve ontological
651 structure, support intervention-consistent reasoning, and expose their own failure modes. LY-GWM
652 demonstrates that explicitly modeling entities, relations, and causal dynamics provides a viable path
653 toward world models that function as reasoning substrates rather than opaque predictors.

654 While the present study focuses on controlled diagnostic environments and a high-fidelity humanoid
655 simulation, the underlying principles are broadly applicable to settings involving long-horizon in-
656 teraction, causal intervention, and autonomous exploration. More generally, these results support
657 the view that world modeling is not solely a problem of representation learning, but a problem of
658 identifying and instantiating the structural conditions under which understanding, intervention, and
659 planning become possible.

660 **8 Conclusion**

661 We introduced LY-GWM, a graph-based world modeling framework designed to support relational
662 reasoning, causal prediction under interventions, goal-directed planning, and novelty-driven explo-

ration. Across controlled diagnostic environments and a high-fidelity humanoid simulation, LY-GWM demonstrates that explicit object-centric structure enables world models to go beyond accurate prediction, functioning instead as interpretable internal simulators for embodied agents.

Our quantitative results show that LY-GWM matches or surpasses strong baselines on reconstruction, causal prediction, goal achievement, and exploration diversity, while qualitative analyses reveal that the model preserves scene structure, responds selectively to interventions, supports planning via rollout, and exposes meaningful novelty signals when encountering out-of-distribution configurations. Together, these findings suggest that structure-aware world models offer a promising foundation for integrating perception, prediction, and control.

Looking forward, we believe that combining graph-based world models with active data collection, adaptive exploration, and uncertainty-aware planning will be critical for scaling to more complex and open-ended environments. By grounding world modeling in explicit relational and causal structure, LY-GWM represents a step toward world models that are not only accurate, but also understandable, diagnosable, and practically useful for embodied intelligence.

A Additional Details of LY-GWM

This appendix provides additional technical details of the LY-GWM framework, including the formal graph ontology, representation choices, training objectives, and implementation details. These details are omitted from the main text for clarity, but are essential for reproducibility and for understanding the design decisions behind the proposed world model.

A.1 Graph Ontology and Scene Representation

LY-GWM represents the environment state at each timestep as a typed, attributed scene graph:

$$G = (V, E, A, R),$$

where:

- $V = \{v_i\}_{i=1}^N$ denotes a set of object-centric nodes;
- $A_i \in \mathbb{R}^{d_a}$ denotes the attribute vector associated with node v_i ;
- $R = \{r_k\}_{k=1}^K$ denotes a finite set of relation types;
- $E \subseteq V \times R \times V$ denotes typed relational edges.

Each node corresponds to a persistent object identity across time. In the diagnostic environments, node attributes include physical state variables such as position, velocity, object type indicators, and optional latent factors. Relations encode pairwise spatial or interaction predicates (e.g., relative position, proximity, or contact), and are represented as multi-channel edge features.

This explicit factorization enforces an object-centric inductive bias and separates entity attributes from relational structure, which is critical for causal reasoning and counterfactual prediction.

A.2 Graph Encoding and Message Passing

At each timestep t , the scene graph G_t is encoded using a graph neural network. Node attributes are first embedded independently:

$$h_i^{(0)} = \phi_{\text{node}}(A_i),$$

where ϕ_{node} is a learnable multilayer perceptron.

Relational messages are computed using both edge attributes and the current action a_t . For each ordered pair (i, j) , a message is formed as:

$$m_{ij} = \phi_{\text{edge}}([E_{ij}, \text{onehot}(a_t)]),$$

where ϕ_{edge} is a shared edge encoder and $\text{onehot}(a_t)$ denotes the discrete action encoding.

Incoming messages are aggregated by summation:

$$m_i = \sum_j m_{ij},$$

703 and used to update node embeddings:

$$h_i^{(1)} = \phi_{\text{update}}([h_i^{(0)}, m_i]).$$

704 The predicted next-state node attributes are obtained via a residual prediction:

$$\hat{A}_i^{t+1} = A_i^t + \phi_{\text{out}}(h_i^{(1)}).$$

705 This formulation encourages the model to learn local, object-level state changes rather than uncon-
706 strained global transitions.

707 A.3 Training Objectives

708 LY-GWM is trained using supervised next-step prediction losses on graph-structured state transi-
709 tions. For a transition (G_t, a_t, G_{t+1}) , the primary loss is:

$$\mathcal{L}_{\text{dyn}} = \sum_{i=1}^N \|\hat{A}_i^{t+1} - A_i^{t+1}\|_2^2.$$

710 When relational attributes are predicted explicitly, an analogous loss is applied to edge features. All
711 losses are averaged over nodes and timesteps.

712 No auxiliary pixel reconstruction or contrastive losses are used in the diagnostic experiments, en-
713 suring that performance reflects the quality of learned relational dynamics rather than perceptual
714 supervision.

715 A.4 Counterfactual Prediction

716 Counterfactual predictions are generated by intervening on the action variable while holding the
717 current graph state fixed. Given a state G_t and an alternative action $a'_t \neq a_t$, the model predicts:

$$\hat{G}_{t+1}^{(a'_t)} = f_{\theta}(G_t, a'_t),$$

718 where f_{θ} denotes the learned causal dynamics.

719 Because actions only enter the model through the edge-level message construction, interventions
720 selectively affect only those objects and relations causally linked to the action, rather than inducing
721 global state shifts.

722 A.5 Model-Based Rollouts and Planning

723 For planning experiments, LY-GWM is rolled out autoregressively over a fixed action sequence:

$$\hat{G}_{t+k+1} = f_{\theta}(\hat{G}_{t+k}, a_{t+k}), \quad k = 0, \dots, H - 1.$$

724 No ground-truth corrections are applied during rollout. The close agreement between imagined
725 and real trajectories observed in Section 6 indicates that the learned dynamics remain stable over
726 multi-step horizons and can serve as an internal simulator for model-based control.

727 A.6 Novelty Estimation via Prediction Error

728 Prediction error is used as an intrinsic signal for novelty detection. At each timestep, a scalar novelty
729 score is computed as the mean-squared prediction error over all node and edge attributes:

$$\mathcal{N}_t = \frac{1}{N} \sum_{i=1}^N \|\hat{A}_i^{t+1} - A_i^{t+1}\|_2^2.$$

730 As shown in Section 6, this signal remains low for in-distribution transitions and increases sharply
731 when the agent encounters previously unseen configurations, making it suitable for exploration pri-
732 oritization or safety monitoring.

733 **A.7 Implementation Details**

734 All models are implemented in PyTorch. Training uses the Adam optimizer with a fixed learning
735 rate. Batch sizes, hidden dimensions, and rollout horizons follow the settings described in Section 5.
736 Experiments are conducted with fixed random seeds, and all reported results are averaged over
737 multiple runs where applicable.